SS 465 DATA FROM KLEINMETALS

REFERENCE LINK

https://www.kleinmetals.ch/shop/datenblatt/E/chronifer-465-kl-usn-s46500-ams-5936-steel-datasheet-515.pdf

Mechanical properties of wires

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Condition	Rm (MPa)	$R_{0.2}$ (MPa)	A_{4d} (%)	Reduction of area (%)	Hardness HRc
Solution annealed	950	770	20	75	29.5
71% cold drawn	1200	1125	12	74	38.5
Solution annealed + aged H900 (482°C)	1779	1703	14	51	50
Solution annealed KV CD + aged H900 (482°C)	2090	2020	10	57	53

Condition	Yield strength $R_{0.2\%}$ L - T (MPa)	UTS Rm L - T (MPa)	Elongation 4d L - T (%)	Contraction* L - T (%)
Solution annealed	683 – 683	951 - 951	20	–
H900 (482°C)	1641 – 1613	1772 – 1772	13 – 12	0.08 – 0.07
H950 (510°C)	1620 – 1586	1751 – 1724	14 – 12	0.11 – 0.10
H1000 (538°C)	1496 – 1455	1593 – 1585	15 – 15	0.14 – 0.13
H1050 (566°C)	1365 – 1351	1482 – 1469	18 – 17	0.16 – 0.16
H1075 (580°C)	1234 – 1241	1400 – 1393	20 – 19	–
H1100 (593°C)	1096 – 1089	1310 – 1310	22 – 21	0.23 – 0.23
H1150M (621°C)	531 – 538	1076 – 1096	25 – 22	0.53 – 0.53

Physical properties

Properties	Units	Condition: Annealed	H900	H1000	H1050	H1100
Density	g cm^{-3}	7.82	7.83	7.85	7.85	7.87
Elastic modulus E	GPa			202.7		199.8
Electrical resistivity	µohm-mm	946	824	822		772
Thermal expansion:	10^{-6} (m m^{-1} K^{-1})					
20 – 100°C		10.30	10.40	10.60		11.30
20 – 200°C		10.80	11.10	11.10		12.00
20 – 400°C		11.10	11.70	11.70		12.70
20 – 600°C		9.86	11.20	12.20		13.10
Thermal conductivity at 23°C	W m^{-1} K^{-1}	14.06	14.85	15.83		15.80
Magnetic properties:						
- Coercivity Hc	Oe	25.5	23.3	28.1	34.2	53.0
- Saturation induction Bs	kG	13.4	13.8	13.3	12.4	10.1

Reference ALLOY Data, Custom 465® Stainless, Carpenter Technology Corporation